

IRELE

Advanced Measurement and Data Driven Modelling

Lab report

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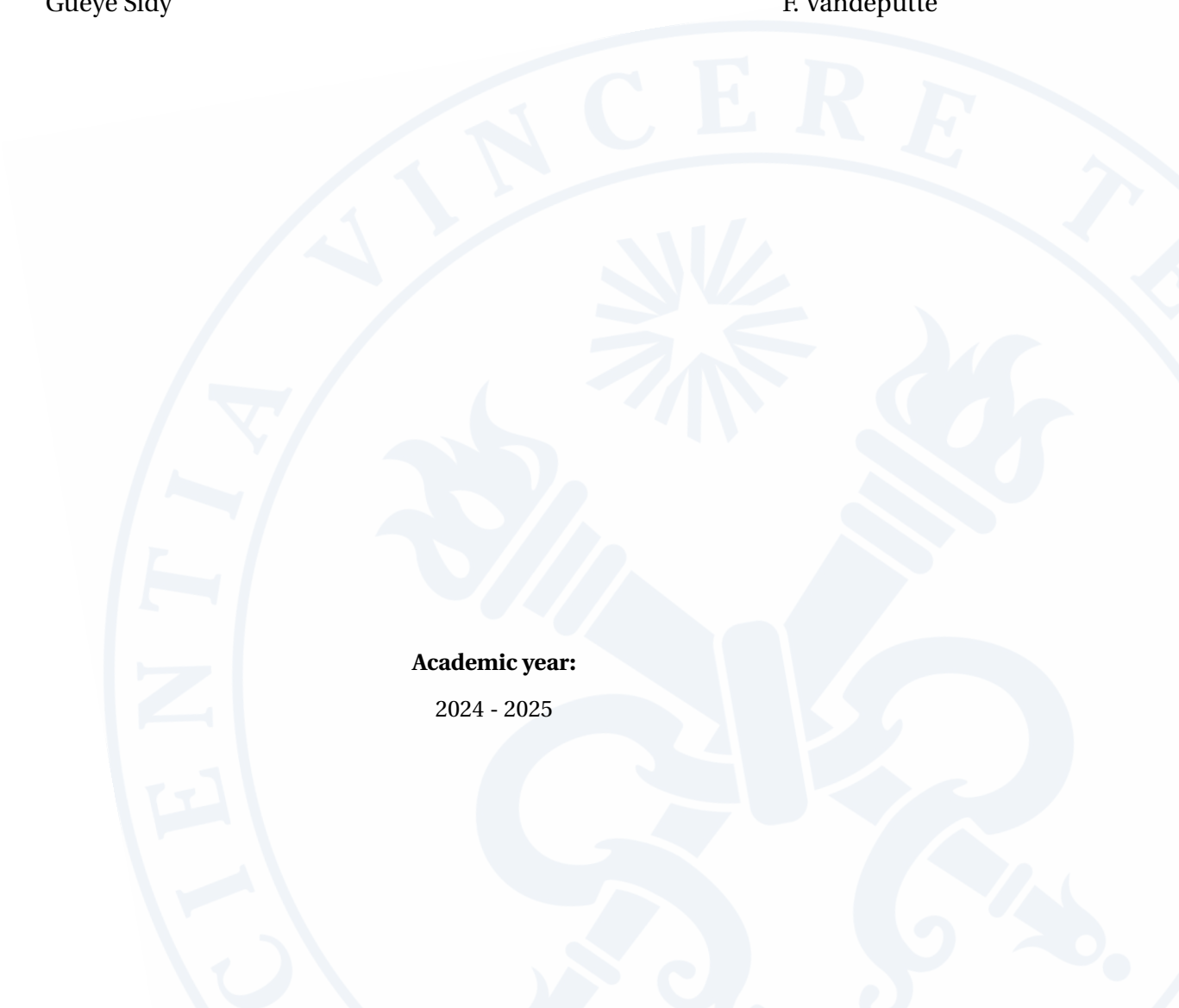
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Lab 1: RLS with forgetting factor

1.1 Main steps

1. Measure the input and output signals from the DUT. The input is arbitrary and the time varying part of the system is due to a manual switch.
2. Make a first estimate of the parameters with an LLS estimator
3. Use the RLS estimator with forgetting factor to track the time-varying parameters of the system.
4. Deduce from the estimated parameters the time instants at which the system changes.

1.2 Time-varying RLS estimation

Starting from the recursive equation of the system, a matrix formulation is derived:

$$y(N) = b_0(N)u(N) + b_1(N)u(N-1) - a_1(N)y(N-1) - a_2(N)y(N-2) \quad (1.1)$$

$$= K(N)\theta(N) \quad (1.2)$$

with

$$K(N) = \begin{bmatrix} u(N) & u(N-1) & -y(N-1) & -y(N-2) \end{bmatrix}, \quad \theta(N) = \begin{bmatrix} b_0(N) \\ b_1(N) \\ a_1(N) \\ a_2(N) \end{bmatrix} \quad (1.3)$$

The RLS estimator (with forgetting factor) is a recursive algorithm that computes the parameters $\theta(N)$ and their covariance matrix K_N after each new sample is collected. Because it is recursive, it needs to be initialized and this is done with a linear least squares estimate on a few first samples.

The linear least squares estimator, used to find θ in $y_{\text{lls}} = H\theta$, is in our case constructed as follows:

$$H = \begin{bmatrix} u(n_{\text{start}} + 1) & u(n_{\text{start}}) & -y(n_{\text{start}}) & -y(n_{\text{start}} - 1) \\ u(n_{\text{start}} + 2) & u(n_{\text{start}} + 1) & -y(n_{\text{start}} + 1) & -y(n_{\text{start}}) \\ \vdots & \vdots & \vdots & \vdots \\ u(n_{\text{LLS}} - 1) & u(n_{\text{LLS}} - 2) & -y(n_{\text{LLS}} - 2) & -y(n_{\text{LLS}} - 3) \end{bmatrix}, \quad y_{\text{lls}} = \begin{bmatrix} y(n_{\text{start}} + 1) \\ y(n_{\text{start}} + 2) \\ \vdots \\ y(n_{\text{LLS}} - 1) \end{bmatrix} \quad (1.4)$$

Where $n_{\text{start}} = \max(n_a, n_b)$ and n_{LLS} is the number of samples used for the LLS estimate, chosen to be 200 here.

$\theta(n_{\text{LLS}})$ and $P_{n_{\text{LLS}}}$ are found with, respectively, the solution to the LLS problem and the definition of the covariance matrix in the RLS algorithm:

$$\theta(n_{\text{LLS}}) = (H^T H)^{-1} H^T y_{\text{lls}}, \quad P_{n_{\text{LLS}}} = (H^T H)^{-1} \quad (1.5)$$

The Matlab code of the LLS initialization is given below:

```
% initialization with an LLS estimate over a few of the first samples
% y_lls = H * theta
H = zeros(sampleForLLS - max(na, nb), na + nb + 1);
for uCol = 0:nb
    H(:, uCol + 1) = u((max(na, nb) + 1 - uCol:sampleForLLS - uCol));
end
for yCol = 1:na
    H(:, nb + 1 + yCol) = -y((max(na, nb) + 1 - yCol:sampleForLLS - yCol));
end
y_lls = y((max(na, nb) + 1:sampleForLLS));

theta = H \ y_lls;
P = inv(H' * H);
```

The recursive least squares equations with forgetting factor g are given by:

$$\theta(N) = \theta(N-1) - \frac{P_{N-1} K^T(N) [K(N) \theta(N-1) - y(N)]}{g + K(N) P_{N-1} K^T(N)} \quad (1.6)$$

$$P_N = \frac{1}{g} \left(P_{N-1} - \frac{P_{N-1} K^T(N) K(N) P_{N-1}}{g + K(N) P_{N-1} K^T(N)} \right) \quad (1.7)$$

The Matlab code implementing these equations is given below:

```
theta_sav = zeros(na + nb + 1, N - max(na, nb) + 2);
P_diag_sav = zeros(na + nb + 1, N - max(na, nb) + 2);

for itr = sampleForLLS+1:N
```

```

% saving previous values
theta_sav(:, itr - max(na, nb) + 1) = theta;
P_diag_sav(:, itr - max(na, nb) + 1) = diag(P);

K_itr = [u(itr - (0:nb)), -y(itr - (1:na))];

theta = theta - (P * (K_itr')) * (K_itr * theta - y(itr)) / (g + K_itr * P * (K_itr'));
P = 1/g * (P - (P * (K_itr')) * K_itr * P) / (g + K_itr * P * (K_itr'));

end

```

1.3 Results

The input of the system is not interesting to show here as it is random white noise. The evolution in time of the estimated parameters is however much more interesting. The following figures will display the evolution of the parameters (with their uncertainty range) for multiple forgetting factor values and for different experiments. The experiments are :

1. Excitation of a LTI sub-system (namely Low LTI) - Figure 1.1.
2. Excitation of the second LTI sub-system (namely High LTI) - Figure 1.2.
3. Excitation of both LTI systems trough a single switch - Figure 1.3.
4. Excitation of both LTI systems trough multiple switches - Figure 1.4.

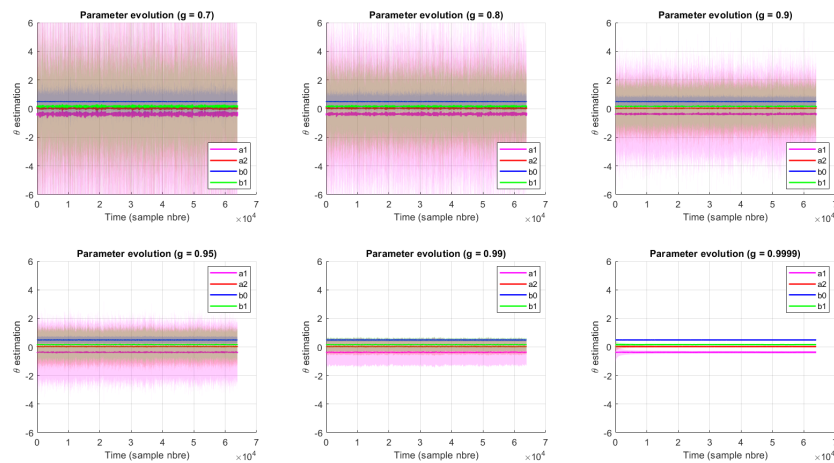


Figure 1.1: Parameters evolution for multiple forgetting factors - Low LTI sub-system

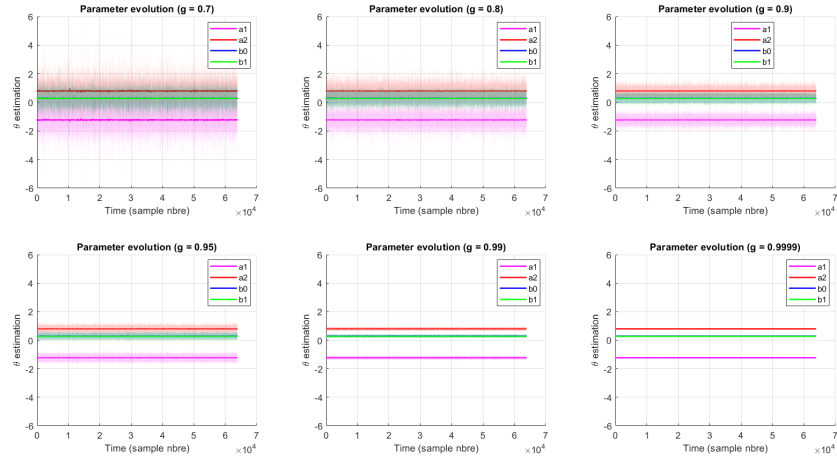


Figure 1.2: Parameters evolution for multiple forgetting factors - High LTI sub-system

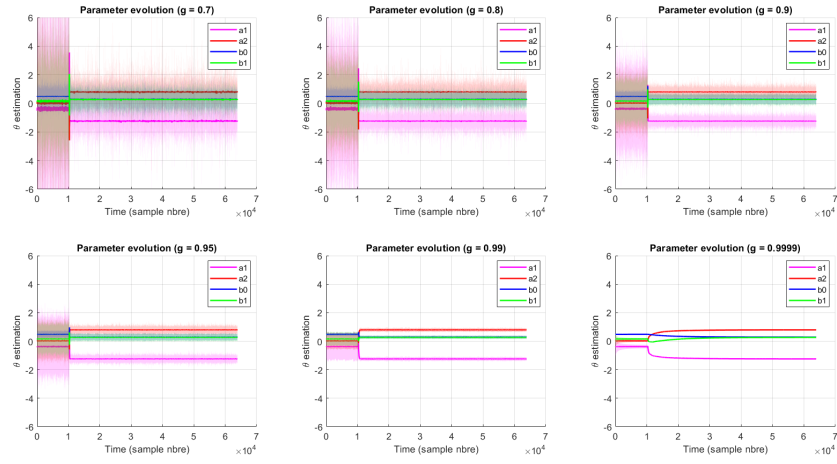


Figure 1.3: Parameters evolution for multiple forgetting factors - single switch between the two LTI systems

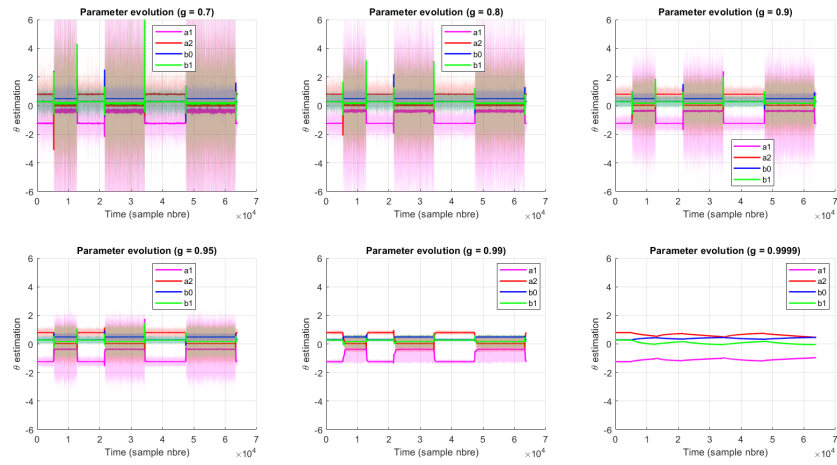


Figure 1.4: Parameters evolution for multiple forgetting factors - multiple switches between the two LTI systems

One may already observe that the uncertainty on the parameters of the LTI system named High LTI system is around 4 times lower than the uncertainty on the parameters for the second LTI system.

Another observation is related to the reduction of the uncertainty converging to a plateau value as the time passes at the beginning of the measurement. This behaviour is well visible in the case of the Low LTI subsystem for a forgetting factor set at 0.9999. The reason behind the uncertainty level diminution is laying in the limited number of samples available for the RLS algorithm at the beginning of the measurements - Figure 1.5.

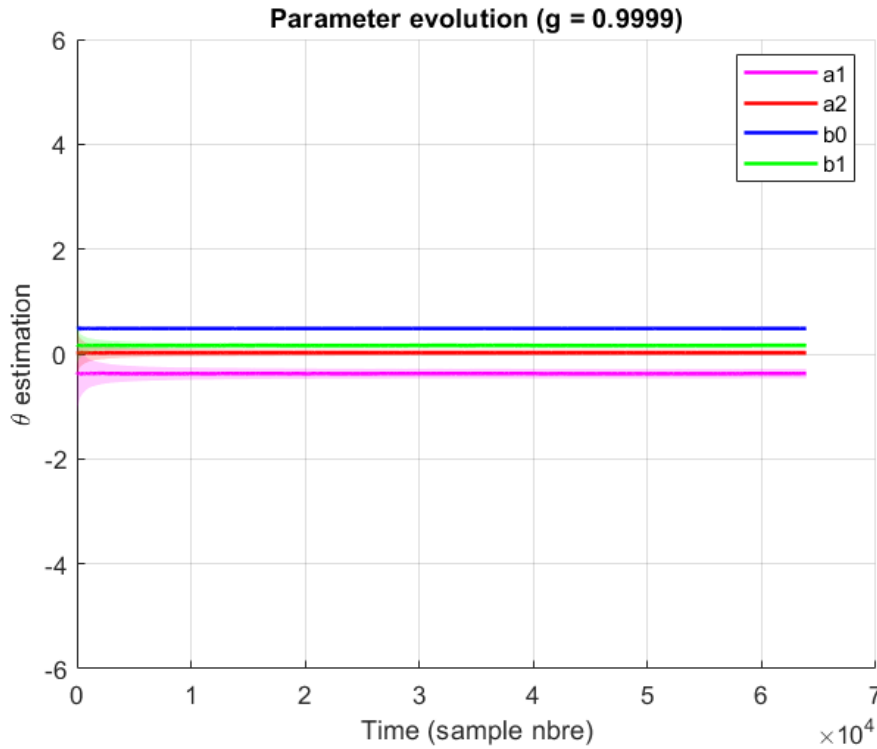


Figure 1.5: Uncertainty variation during the first 500 samples - Low LTI system ($g=0.9999$)

As shown on the figures, the RLS algorithm with forgetting factor is a powerful tool to track the parameters of time-varying system simulated in our case by switching two LTI subsystems. Nevertheless, the choice of the forgetting factor value is dependent of the constraints or specifications of our system and also the required confidence on the parameters we get once in steady-state.

Let's consider our experiment with multiple switches - One can easily see that the RLS algorithm with a forgetting factor of 0.9999 cannot converge fast enough and is therefore not relevant for such time-varying system.

In the other hand, the other RLS algorithm with forgetting factor smaller than 0.9999 exhibits higher parameter variances leading to uncertainty much more significant (10 times higher in case of $g = 0.99$); but also noticeable spikes at the switching instants resulting to a non-smooth evolution of the parameters. This rough evolution may result to potential issues, damages in real systems we are tracking (in case by example

of RLS algorithm used for control system). For rigorosity, a table containing the parameters variance for the different forgetting factors is shown below - Table 1.1 and 1.2.

Low LTI	σ_{a1}^2	σ_{a2}^2	σ_{b0}^2	σ_{b1}^2
g=0.7	46.0729	19.1852	0.3077	13.5596
g=0.8	19.5318	8.1543	0.1653	5.5176
g=0.9	7.2197	2.7688	0.0767	1.9273
g=0.95	3.5704	1.2898	0.0393	0.9333
g=0.99	0.2875	0.1140	0.0090	0.0791
g=0.9999	$0.2385e^{-3}$	$0.2123e^{-3}$	$0.1125e^{-3}$	$0.1429e^{-3}$

Table 1.1: Parameters variances in steady state for Low LTI system

High LTI	σ_{a1}^2	σ_{a2}^2	σ_{b0}^2	σ_{b1}^2
g=0.7	1.2416	1.4443	2.0137	2.7133
g=0.8	0.5458	0.6222	0.7531	0.9635
g=0.9	0.1918	0.1995	0.2014	0.2356
g=0.95	0.0874	0.0842	0.0694	0.0798
g=0.99	0.0204	0.0189	0.0117	0.0135
g=0.9999	$0.4765e^{-3}$	$0.3578e^{-3}$	$0.1062e^{-3}$	$0.1921e^{-3}$

Table 1.2: Parameters variances in steady state for High LTI system

It remains therefore to decide the most suitable forgetting factor for our experiment with multiple switches. In our case, the forgetting factor of 0.99 was converging few hundreds of samples after the switch (0.15 seconds) to its steady-state - fast enough to converge before the next switching and with a smooth convergence. The switching instants and the convergence of the parameters to its steady-state is shown at the figure 1.6.

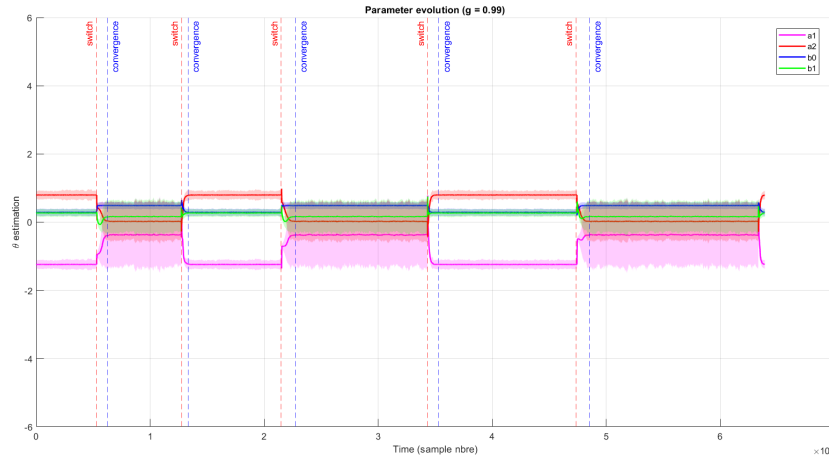


Figure 1.6: Switching instants and convergence (g=0.99)

In order to grasp the differences between the results given for different forgetting factor, it is useful to plot the transfer function modeled from the parameters given by the RLS algorithms and compared it at the FRF computed with the average power spectrum method from the input signal and the output measurements. The timing to compare the models and the FRF is chosen once the RLS algorithm with g=0.99 has

reached its steady-state after a switch of system - Figure .

We can easily see that RLS algorithm with forgetting factor smaller than 0.99 have reached the convergence and match the estimated FRF (which is logical as those algorithm are reacting faster than the algorithm with $g=0.99$). Nevertheless, by looking carefully to the modeled transfer functions coming from RLS with forgetting factors smaller than 0.99, one can observe the models are fitting the FRF with a lower accuracy resulting from the higher uncertainty level on the parameters.

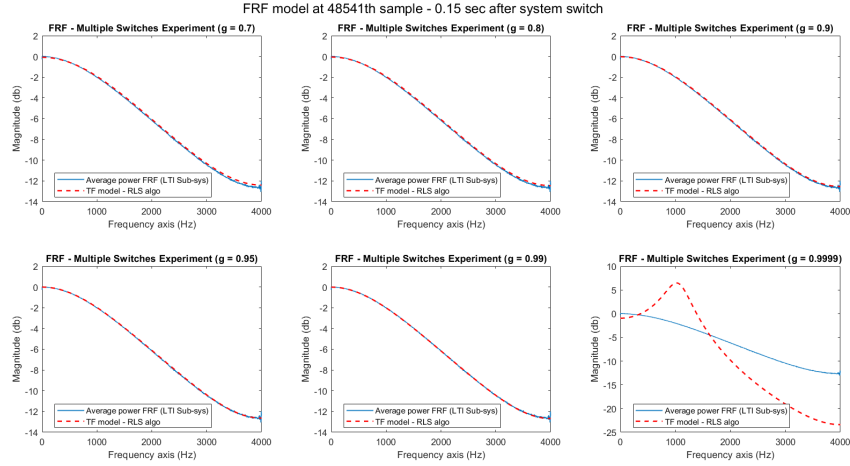


Figure 1.7: Comparison estimated FRF and computed model - at steady state instant for RLS with forgetting factor set at 0.99

Finally, the comparison between the estimated FRF for both LTI sub-systems and the transfer function calculated based on the parameters - computed by the RLS algorithm with forgetting factor of 0.99 once the steady-state is reached - are displayed at the figure 1.8 and 1.9

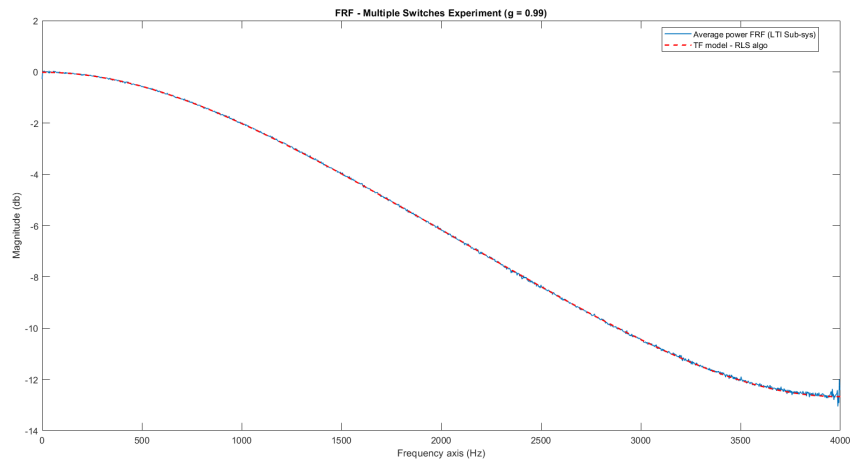


Figure 1.8: Comparison estimated FRF and computed model - Low LTI system - $g=0.99$

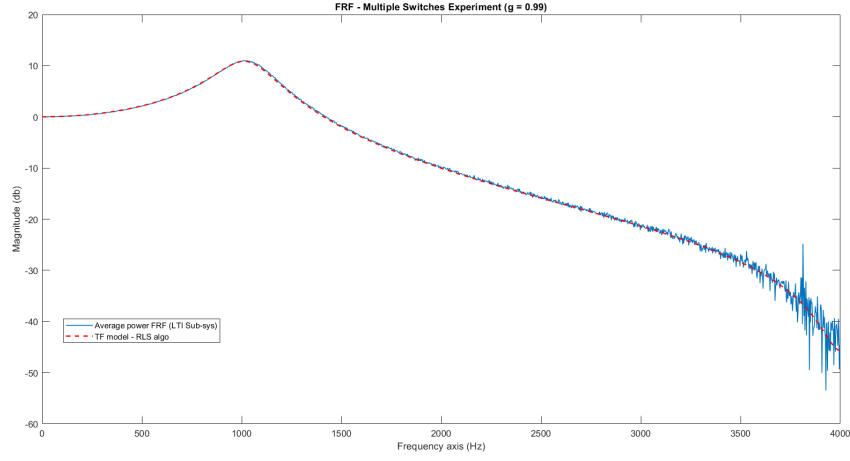


Figure 1.9: Comparison estimated FRF and computed model - High LTI system - $g=0.99$

Finally, the mathematical expression of the transfer functions (RLS - $g=0.99$ - steady-state instants) are detailed below:

$$G_{low}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.4874 + 0.1613 z^{-1}}{1 - 0.3763 z^{-1} + 0.0263 z^{-2}} \quad (1.8)$$

$$G_{high}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.2871 + 0.2713 z^{-1}}{1 - 1.2378 z^{-1} + 0.7951 z^{-2}} \quad (1.9)$$

In comparison the transfer function computed from LLS for each LTI sub-system taken separately :

$$G_{low-LLS}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.4832 + 0.1752 z^{-1}}{1 - 0.3518 z^{-1} + 0.01 z^{-2}} \quad (1.10)$$

$$G_{high-LLS}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.2884 + 0.2659 z^{-1}}{1 - 1.2419 z^{-1} + 0.7964 z^{-2}} \quad (1.11)$$

1.4 Conclusion

See then

2.1 Main steps

1. TODO

2.2 State space model

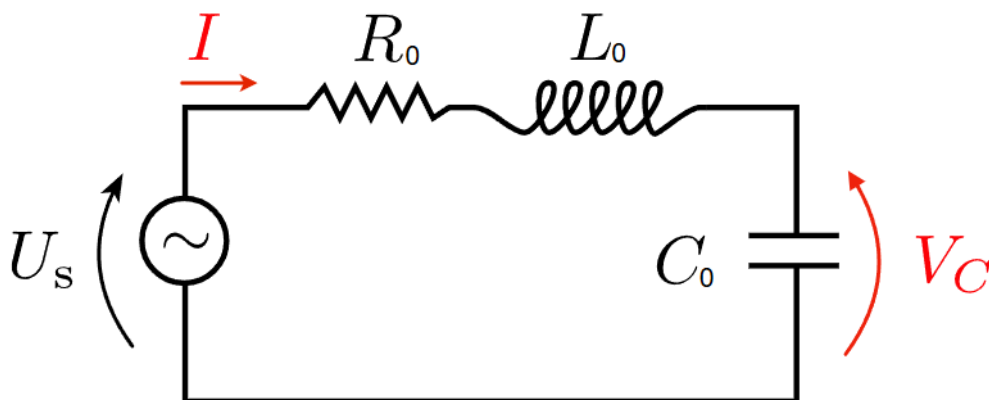


Figure 2.1: Studied RLC circuit

The studied system is the RLC circuit shown in Fig 2.1. A set of equations can be derived using Kirchhoff's laws:

$$u_s(t) = R_0 i(t) + L_0 \frac{di(t)}{dt} + \frac{1}{C_0} \int i(t) dt \quad (2.1)$$

$$v_C(t) = \frac{1}{C_0} \int i(t) dt \quad (2.2)$$

By substituting the second equation into the first one, we obtain:

$$u_s(t) = R_0 i(t) + L_0 \frac{di(t)}{dt} + v_C(t) \quad (2.3)$$

To get to the continuous-time state space representation of this circuit, we need to define the input of the system as $u_s(t)$, its output as $v_C(t)$ and the state vector as:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} v_C(t) \\ i(t) \end{bmatrix} \quad (2.4)$$

Which finally leads to:

$$\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{C_0} \\ -\frac{1}{L_0} & -\frac{R_0}{L_0} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L_0} \end{bmatrix} u_s(t) \quad (2.5)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \quad (2.6)$$

$$\Leftrightarrow \quad A_c = \begin{bmatrix} 0 & \frac{1}{C_0} \\ -\frac{1}{L_0} & -\frac{R_0}{L_0} \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ \frac{1}{L_0} \end{bmatrix}, \quad C_c = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_c = 0 \quad (2.7)$$